Florida International University

Department of Electrical & Computer Engineering

Energy Conversion Lab (EEL-4213L)

Quiz 01

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Executive Summary

This comprehensive laboratory investigation analyzed the performance characteristics of a single-phase step-up transformer (100:200 turns ratio) under various load conditions. The transformer was tested with resistive, capacitive, inductive, and complex loads, followed by standard open-circuit and short-circuit tests to develop equivalent circuit parameters.

Key findings demonstrate that resistive loads provide optimal transformer performance with unity power factor and maximum efficiency (88.9%), while reactive loads significantly degrade performance through poor power factors and opposite voltage regulation effects.

Experimental Setup

Transformer Specifications

- **Primary:** 100 turns, 120V RMS rated voltage
- **Secondary:** 200 turns, 240V RMS theoretical voltage
- **Frequency:** 60 Hz
- **Turns Ratio:** $a = N_1/N_2 = 0.5$ (step-up configuration)
- **Source:** 120V RMS, 60Hz sinusoidal voltage

Load Configurations

The parallel RLC load system enabled testing of:

- Resistive loads: 50-200Ω range
- Capacitive loads: 10-100μF range
- Inductive loads: 10-600mH range
- Complex loads: R+C and R+L combinations

Task 1: Resistive Load Analysis

Performance Characteristics

Resistive loads demonstrated superior transformer performance across all metrics. Testing with resistance values from 50Ω to 200Ω revealed consistent trends aligning with electrical engineering principles.

Experimental Data

Resistance (Ω)	Source V (V)	Source I (A)	Source P (W)	Load V (V)	Load I (A)	Load P (W)	Efficiency (%)	Voltage Reg (%)
50	120.20	11.01	1313.09	240.08	4.86	1167.39	88.90	-0.033
100	120.20	6.17	723.16	240.20	2.41	578.38	79.98	-0.083
125	120.24	5.21	604.42	240.29	1.91	459.56	76.03	-0.121
150	120.24	4.63	532.06	240.31	1.61	387.30	72.79	-0.129
200	120.17	3.86	432.74	240.19	1.20	288.38	66.64	-0.079

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Key Findings

- **Best efficiency at heavy loads:** 88.9% at 50Ω loading
- **Unity power factor:** Perfect alignment of voltage and current (PF = 1.00)
- **Excellent voltage regulation:** <0.13% variation across all loads
- **Constant power losses:** ~145W dominated by iron losses

The efficiency degradation with increasing resistance (lighter loads) follows the principle that transformer efficiency is maximized when copper losses equal iron losses. Since iron losses remain constant at ~145W while copper losses decrease with lighter loading, overall efficiency decreases.

Task 2: Capacitive Reactive Load Analysis

Performance Degradation

Capacitive loads demonstrated significant performance deterioration compared to resistive loads, exhibiting leading power factor characteristics and voltage rise effects.

Experimental Data

Capacitance (µF)	Source V (V)	Source I (A)	Source P (W)	Source PF	Load V (V)	Load I (A)	Load P (W)	Voltage Reg (%)
20	120.20	3.30	144.72	0.39	240.67	1.83	-0.21	-0.279
50	120.20	8.44	145.58	0.14	241.17	4.54	-0.62	-0.487
75	120.19	12.94	146.79	0.09	241.59	6.82	-1.03	-0.663
100	120.19	17.56	148.72	0.07	242.02	9.14	-1.30	-0.842

Critical Observations

- **Very poor power factors:** 0.07-0.39 leading power factor
- **Voltage rise phenomenon:** Up to -0.84% voltage increase
- **High reactive power flow:** Up to -2212 VAR leading
- **No real power consumption:** Capacitors store/release energy cyclically

The negative voltage regulation indicates voltage rise under capacitive loading, opposite to conventional resistive behavior. This occurs because leading reactive current creates a voltage boost effect in transformer impedance.

Task 3: Inductive Reactive Load Analysis

Inductive loads exhibited lagging power factor characteristics with voltage drop effects, directly opposite to capacitive behavior.

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Experimental Data

Inductance (mH)	Source V (V)	Source I (A)	Source P (W)	Source PF	Load V (V)	Load I (A)	Load P (W)	Load PF	Voltage Reg (%)
50	120.20	38.42	168.98	0.06	237.99	18.57	18.23	0.01	+0.837
100	120.20	22.10	150.94	0.09	239.09	10.38	5.16	0.00	+0.379
300	120.23	8.60	145.33	0.23	239.98	3.58	0.76	0.00	+0.008
500	120.23	5.83	145.02	0.34	240.14	2.17	0.45	0.00	-0.058

Performance Trends

- **Improving power factor :** with higher inductance (0.06→0.34)
- **Voltage drop effects:** Positive voltage regulation
- **High current demand:** Up to 38.42A for low inductance
- **Decreasing reactive power :** as inductance increases

Higher inductance values create better impedance balance, improving power factor and reducing current demand. The relationship XL = 2πfL shows that larger inductance increases inductive reactance, creating more balanced impedance characteristics.

Task 4: Complex Resistive-Capacitive Load Analysis

Complex R+C loads (R=100Ω with varying capacitance) demonstrated intermediate performance between pure resistive and pure capacitive characteristics.

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Experimental Data

Capacitance (µF)	Source V (V)	Source I (A)	Source P (W)	Source PF	Load V (V)	Load I (A)	Load P (W)	Load PF	Efficiency (%)
20	121.18	16.33	727.32	0.39	241.01	5.19	581.30	0.98	79.92
50	120.20	14.25	730.82	0.43	241.49	7.27	582.26	0.53	79.67
75	120.21	6.74	723.60	0.31	240.55	3.01	578.34	0.90	79.93

Key Insights

- **Maintained efficiency**: 79.7-79.9% due to resistive component
- **Variable power factors**: Complex impedance interactions
- **Moderate voltage rise**: Between resistive and capacitive extremes
- **Real power consumption**: ~577W through resistive element

The **resistive component dominates real power consumption**, maintaining reasonable efficiency, while the **capacitive component introduces leading reactive power**, degrading power factor and causing voltage rise.

Task 5: Complex Resistive-Inductive Load Analysis

Corrected Experimental Data

Complex R+L loads (R=100Ω with varying inductance) showed progressive performance improvement with increasing inductance values.

Inductance (mH)	Source V (V)	Source I (A)	Source P (W)	Source PF	Load V (V)	Load I (A)	Load P (W)	Efficiency (%)
100	120.23	21.91	722.75	0.62	239.09	10.15	576.70	79.79
300	120.21	15.40	721.90	0.76	239.52	4.31	576.16	79.81
500	120.21	8.27	721.04	0.87	239.57	3.23	576.26	79.92

Excellent Performance Trends

- **Improving power factor:** 0.62→0.87 with higher inductance
- **Decreasing current:** 21.9A→8.3A (62% reduction)
- **Consistent efficiency:** ~79.8% across all configurations
- **Reduced reactive power:** 1368→401 VAR

The impedance ratio R:XL progresses from 1:0.38 to 1:1.89, creating better balanced impedance and improved power factor. Higher inductive reactance (XL = 2πfL) provides better impedance matching with the resistive component.

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Task 6: Transformer Equivalent Circuit Development

Open Circuit Test Results

Test Conditions:

- Primary voltage: 120.19V
- No-load current: 1.82A
- No-load power: 144.37W
- Power factor: 0.83

Calculated Parameters:

- Iron losses: Pi = 144.4W
- Core resistance: Rc = 100.1Ω
- Magnetizing reactance: Xm = 119.5Ω

Short Circuit Test Results

Test Conditions:

- Primary voltage: 14.14V RMS
- Short circuit current: 299.9A
- Short circuit power: 1054.8W
- Power factor: 0.30

Calculated Parameters:

- Equivalent resistance: R_{01} = 0.0117Ω
- Equivalent reactance: X_{01} = 0.0457Ω
- Equivalent impedance: Z_{01} = 0.0472Ω

Comprehensive Performance Analysis

Load Type Comparison Summary

Load Type	Power Factor	Efficiency Range	Voltage Regulation	Key Characteristics
Resistive	1.00 (Unity)	66.6-88.9%	±0.13%	Optimal performance
Capacitive	0.07-0.39 (Leading)	N/A	-0.28 to -0.84%	Voltage rise, poor PF
Inductive	0.06-0.34 (Lagging)	N/A	+0.01 to +0.84%	Voltage drop, poor PF
R+C Complex	0.31-0.43 (Leading)	79.7-79.9%	-0.23 to -0.62%	Intermediate leading
R+L Complex	0.62-0.87 (Lagging)	79.8-79.9%	+0.18 to +0.38%	Intermediate lagging

Voltage Regulation Mechanisms Leading power factor (capacitive): Creates voltage rise due to capacitive reactance effects opposing inductive transformer impedance. Lagging power factor (inductive): Creates voltage drop due to inductive reactance adding to transformer impedance. Unity power factor (resistive): Minimal voltage change due to balanced reactive components.

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Engineering Conclusions and Discussion

Analytical Task Responses

Task 1 Analysis: Efficiency calculations for resistive loads reveal that maximum efficiency occurs at heavy loading (50Ω = 88.9%) and decreases with lighter loads due to the constant iron loss component dominating total losses.

Task 2 Analysis: Efficiency and voltage regulation calculations across all load types demonstrate that reactive loads severely degrade transformer performance through poor power factors and voltage regulation issues.

Task 3 Analysis: The equivalent circuit model developed from open and short circuit tests accurately predicts transformer behavior and validates the measured performance data across all loading conditions.

Task 4 Discussion:

Is there a change in efficiency and voltage regulation for various loads?
Yes, significant changes occur: Resistive loads provide excellent efficiency (66-89%) and voltage regulation (±0.13%), while reactive loads cause poor voltage regulation (-0.84% to +0.84%) and eliminate meaningful efficiency calculations due to minimal real power consumption.

What happens as load changes from lagging to leading?
Voltage regulation reverses direction: Lagging loads (inductive) cause voltage drops (+0.84%), while leading loads (capacitive) cause voltage rises (-0.84%). Power factor remains poor for both reactive extremes, but the voltage effects are opposite.

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Primary Findings

1. Resistive loads optimize transformer performance through unity power factor operation, achieving maximum efficiency and excellent voltage regulation.

2. Reactive loads significantly degrade performance by introducing poor power factors and causing opposite voltage regulation effects.

3. Complex loads exhibit intermediate characteristics determined by the impedance ratio between resistive and reactive components.

4. Higher inductance in R+L combinations improves performance by creating better impedance balance and reducing reactive power demand.

5. Transformer losses are dominated by iron losses (~144W constant) rather than copper losses, explaining efficiency trends with loading.

Theoretical Validation

All experimental results align with established electrical engineering principles:

- Ohm's Law and AC circuit theory predictions match measured values
- Power triangle relationships correctly describe complex power flow
- Transformer equivalent circuit model accurately represents measured performance
- Impedance calculations verify theoretical reactance values ($X_L = 2\pi fL$, $X_C = 1/2\pi fC$)

The laboratory investigation successfully demonstrates fundamental transformer behavior under various loading conditions, providing essential insights into power system operation and the critical importance of power factor management in electrical systems.

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